

ASSESSMENT

Q1. ResNet Skip Connections in Production

Topic: ResNet & Skip Connections	Difficulty: Hard	Points: 7 marks
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SCENARIO

Your team is training a 152-layer ResNet for medical image classification. Training loss is decreasing steadily, but validation accuracy plateaus at epoch 12 and begins degrading. A junior engineer suggests removing some residual connections to "reduce complexity." Meanwhile, the gradient norms in early layers are orders of magnitude smaller than in final layers.

TASK

Identify what is happening architecturally, explain why the junior engineer's suggestion would worsen — not fix — the problem, and describe two concrete interventions you would apply at the architecture or training level to resolve this.

Hint: Consider what skip connections specifically solve and what happens to gradients in their absence.

The issue described is not due to the presence of skip connections but rather due to overfitting and instability in training. The model is learning the training data very well as seen from the decreasing training loss but its ability to generalize is getting worse after a certain point which is why the validation accuracy starts dropping after epoch twelve.

At the same time increasing gradient norms in deeper layers indicate that gradients are becoming unstable and possibly growing too large. Skip connections in a deep residual network are designed to allow gradients to flow easily across layers and prevent vanishing gradient problems.

Removing them would break this direct path and make training a very deep network much harder which could lead to worse performance instead of improvement. The correct approach is to address overfitting and training instability.

One effective solution is to apply regularization techniques such as adding dropout increasing weight decay and using stronger data augmentation especially because medical datasets are often limited. Another useful intervention is to stabilize training by reducing the learning rate after the point where validation accuracy starts degrading or by using learning rate scheduling along with gradient clipping. Early stopping can also be applied to prevent the model from learning noise. Therefore the architecture should be preserved and the training strategy should be improved.

Q2. Inception vs. Plain Conv in Satellite Imagery

Topic: Inception Networks

Difficulty: Hard

Points: 7 marks

SCENARIO

A geospatial analytics company processes 1m-resolution satellite tiles (512×512 pixels) to detect objects ranging from individual cars (~3×3 pixels) to entire warehouses (~80×80 pixels). Their baseline plain-CNN achieves 68% mAP. A colleague recommends switching to an Inception-style architecture. Budget is constrained: the model must run inference on a 4GB VRAM GPU.

TASK

Explain the specific architectural mechanism in Inception that addresses the multi-scale detection challenge here. Then describe how 1×1 convolutions enable this without exceeding the VRAM budget, using approximate parameter-count reasoning.

Hint: Think about what happens when you try to detect a 3px car and an 80px warehouse with the same fixed kernel.

The main difficulty arises from the fact that objects in satellite images appear at very different scales. A plain convolutional neural network uses fixed size kernels which limits its ability to detect both very small and very large objects effectively.

For example a small car occupying only a few pixels requires fine local features while a large warehouse requires a much broader receptive field. An inception style architecture addresses this by processing the input through multiple convolutional filters of different sizes in parallel so that features at multiple scales are captured simultaneously.

A key component that makes this efficient is the use of one by one convolutions before applying larger filters. These one by one convolutions reduce the number of channels in the feature map which significantly decreases the number of computations required for subsequent larger convolutions.

Without this reduction applying a large filter on a high dimensional input would be computationally expensive and would exceed the available memory. By first compressing the channel dimension and then applying larger filters the model can achieve multi scale feature extraction while still fitting within the memory limits of a four gigabyte GPU. This allows the model to detect both small and large objects more effectively than a plain convolutional network.

Q3. EfficientNet Compound Scaling Decision

Topic: EfficientNet & NAS	Difficulty: Medium	Points: 6 marks
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SCENARIO

You are deploying a product defect classifier on a manufacturing line. You have two hardware targets: (A) a cloud GPU with ample memory, and (B) an edge Raspberry Pi 4 with 4GB RAM and no GPU. Your dataset has 50,000 images at 224×224. You are choosing between EfficientNet-B0, B4, and B7.

TASK

Explain EfficientNet's compound scaling principle and justify which variant you'd deploy to each hardware target. Describe one specific failure mode that can occur when you apply depth scaling without corresponding resolution scaling.

Hint: The compound coefficient ϕ jointly scales width, depth, and resolution — but those dimensions are not independent.

EfficientNet introduces a compound scaling method which balances the scaling of depth, width, and input resolution instead of increasing them independently. This is important because these dimensions are interdependent and scaling only one of them leads to suboptimal performance.

When deploying on a cloud GPU with sufficient memory such as eight gigabytes, a mid-range variant like EfficientNet-B4 is suitable because it offers higher resolution and more capacity to capture fine details, which is important for detecting defects accurately.

On the other hand, when deploying on a resource-constrained device such as a Raspberry Pi with limited memory and processing power, a lightweight model like EfficientNet-B0 is more appropriate because it provides reasonable accuracy while maintaining low computational cost and fast inference.

A common failure occurs when only depth is increased without adjusting width and resolution. In such a case, the network becomes deeper but does not receive richer input information or sufficient channel capacity to learn useful features. This can lead to diminishing returns and even training instability. Therefore, the balanced scaling approach of EfficientNet ensures that improvements in one dimension are supported by proportional changes in the others, resulting in better overall performance.

Q4. MobileNet on a Low-Power Wearable

Topic: MobileNet & SqueezeNet

Difficulty: Medium

Points: 6 marks

SCENARIO

You are building a fall-detection model for a smartwatch running ARM Cortex-M7 at 216 MHz with 512KB RAM. The model must classify 5-frame accelerometer+camera bursts in under 30ms. Standard MobileNetV1 still has 4.2M parameters and is too large.

TASK

Explain depthwise separable convolutions mathematically (show the parameter reduction factor vs. standard convolution for a $D \times D$ kernel with C_{in} and C_{out} channels). Then name two additional compression techniques beyond depthwise separability that you would apply to hit the RAM constraint.

Hint: Write out the parameter counts for a standard 3×3 conv block vs. its depthwise separable equivalent.

The main problem is that a standard convolution layer is too heavy for a low power wearable device with strict memory limits. In a normal three by three convolution with input channels and output channels the number of parameters is calculated as three multiplied by three multiplied by input channels multiplied by output channels. This becomes large when the number of channels increases.

Depthwise separable convolution reduces this cost by splitting the operation into two parts. First a depthwise convolution is applied where each input channel is convolved separately with its own three by three filter so the number of parameters becomes three multiplied by three multiplied by input channels.

After that a pointwise convolution using one by one filters combines these channels to produce the desired number of output channels and this requires input channels multiplied by output channels parameters.

When combined the total number of parameters becomes three multiplied by three multiplied by input channels plus input channels multiplied by output channels which is much smaller than the standard convolution. This reduction makes the model suitable for devices with limited memory.

To further reduce memory usage techniques such as reducing the width of the network by using a width multiplier and applying quantization to use lower precision weights can be used. Pruning unnecessary connections can also help in fitting the model within the available memory while maintaining acceptable performance.

Q5. DenseNet Feature Reuse in Limited Data

Topic: DenseNet	Difficulty: Medium	Points: 6 marks
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SCENARIO

A radiology startup has 1,200 labeled chest X-rays to train a pathology detector. A data scientist argues for DenseNet over ResNet because 'DenseNet uses fewer parameters.' The engineering lead disagrees, claiming ResNet is more memory-efficient at inference time. Both are partially right — but for different reasons.

TASK

Reconcile this debate: explain why DenseNet can use fewer parameters for equivalent accuracy AND why ResNet has lower peak GPU memory at inference. Then describe the specific mechanism in DenseNet that makes it particularly robust in low-data regimes like this one.

Hint: Consider what happens to intermediate feature maps in a dense block during the forward pass.

Both viewpoints are partially correct but they focus on different aspects of efficiency. Densenet is designed to reuse features by connecting each layer to every previous layer which means that instead of relearning similar features the network directly accesses earlier computed feature maps.

This reduces the need for a large number of filters in each layer and therefore the total number of parameters can be lower for similar accuracy especially in small datasets where feature reuse is beneficial.

On the other hand resnet is more memory efficient during inference because it uses additive skip connections rather than concatenation.

In densenet the feature maps from all previous layers are concatenated and passed forward which increases the number of feature maps that need to be stored in memory at any given time. This leads to higher memory consumption during inference even though the number of parameters is smaller.

The key mechanism that allows densenet to perform well on limited data is this dense connectivity which encourages feature reuse and improves gradient flow. Each layer receives direct supervision from earlier layers and does not need to learn redundant representations. However this advantage comes at the cost of increased memory usage due to storing multiple intermediate feature maps.

Q6. Why Faster R-CNN Replaced R-CNN

Topic: R-CNN Family

Difficulty: Hard

Points: 7 marks

SCENARIO

A video surveillance company was using R-CNN to flag suspicious activity in retail stores, processing 1 frame per 47 seconds. They upgraded to Fast R-CNN, which reduced it to 2.3 seconds per frame. A consultant now claims Faster R-CNN will bring it to near real-time. The engineering team is skeptical about the bottleneck analysis.

TASK

Identify the exact computational bottleneck in R-CNN that Fast R-CNN addressed, and then the remaining bottleneck that Fast R-CNN did NOT solve (that Faster R-CNN resolves). Describe the Region Proposal Network and explain why sharing convolutional features between RPN and detection head is architecturally significant.

Hint: Trace where the compute time goes in each model generation.

Faster r-cnn improved over rcnn mainly by removing redundant computation. In rcnn each region proposal is processed independently by running a convolutional network which results in a large amount of repeated computation and makes it very slow.

Fast rcnn addressed this by running the convolutional layers only once on the entire image and then extracting features for each region proposal from the shared feature map using a region of interest pooling operation. This significantly reduced computation time.

Faster rcnn further improved the pipeline by introducing a region proposal network which generates proposals directly from the shared convolutional feature maps instead of relying on an external algorithm. This removed the slow selective search step used in earlier methods.

However even after these improvements faster rcnn is not fully real time because it still processes proposals sequentially in the detection head and the number of proposals can be large which adds latency. The region proposal network is important because it shares convolutional features with the detection network which avoids recomputing features and makes the system more efficient.

This architectural change reduces the overall computation and allows faster rcnn to achieve much better speed compared to earlier approaches while maintaining high accuracy although it still does not reach the speed of single stage detectors.

Q7. YOLO Grid Cell Failure on Dense Crowds

Topic: YOLO	Difficulty: Hard	Points: 7 marks
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SCENARIO

A sports analytics firm uses YOLOv5 to track players in stadium footage. Detection mAP is 91% in open-play but drops to 61% during set pieces where 5–8 players cluster in a 3m radius. The team's first instinct is to collect more training data of clustered players.

TASK

Explain the fundamental architectural reason (not a data issue) why YOLOv5's grid cell design causes this failure. Describe how YOLOv8 or anchor-free YOLO variants address this structurally. Would collecting more data actually fix the root cause? Justify your answer.

Hint: One grid cell in YOLO can predict only N objects. What happens when N players fall in the same cell?

The drop in detection performance in dense crowd situations is mainly due to the way the yolo model divides the image into a fixed grid and assigns responsibility for object prediction.

In this design each grid cell is responsible for predicting a limited number of objects and typically assigns predictions based on the center of the object falling inside that cell. When multiple players are very close together and fall within the same grid cell the model cannot represent all of them effectively because it has a limited number of prediction slots.

As a result some players are ignored which leads to missed detections and a drop in average precision. This is not a data issue but a structural limitation of the grid based detection mechanism. Increasing training data will not fix the fundamental constraint that a single grid cell can only predict a small number of objects.

Anchor based improvements in later yolo versions increase the number of possible predictions per cell by using multiple predefined shapes which helps but still does not fully solve the issue in extremely dense scenarios.

Anchor free approaches remove the dependence on fixed anchor boxes and instead predict object centers or keypoints more flexibly which allows better handling of overlapping objects. Another solution is to increase the resolution of the feature map so that objects are distributed across more cells which reduces collisions within a single cell.

Therefore the failure arises from limited prediction capacity per grid cell and the solution lies in architectural changes rather than simply adding more data.

Q8. SSD Multi-Scale Feature Map Tradeoffs

Topic: SSD	Difficulty: Medium	Points: 6 marks
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SCENARIO

A robotics team is using SSD300 to detect table-top items (cups, tools, small bolts) for a robotic arm. They notice bolts (~5px in the 300×300 input) are almost never detected, while cups (~60px) are detected reliably. A teammate says 'SSD handles multi-scale by design — this shouldn't happen.'

TASK

Explain why very small objects systematically fail in SSD despite its multi-scale feature maps. Describe the specific feature map layer in SSD300 responsible for small objects and what spatial resolution it has. Propose one architectural fix.

Hint: Count the spatial dimensions of each SSD prediction head and ask which is responsible for tiny objects.

Although SSD is designed to detect objects at multiple scales it still struggles with very small objects because of the way its prediction layers are structured. SSD uses multiple feature maps at different resolutions to detect objects of varying sizes.

The earlier layers in the network have higher spatial resolution and are responsible for detecting small objects while deeper layers have lower resolution and are used for larger objects. However even the highest resolution feature map in SSD three hundred is relatively coarse compared to the input image.

When the input is reduced to three hundred by three hundred pixels small objects such as bolts occupy very few pixels and after passing through several convolution and pooling layers their representation becomes too weak or disappears entirely.

This makes it difficult for the detector to localize and classify them accurately. The layer responsible for small object detection has the largest spatial dimensions among the prediction layers but it is still not fine grained enough for tiny objects.

This is why small items are often missed despite the multi scale design. The limitation is therefore due to loss of spatial detail and insufficient resolution in early feature maps. Improvements can be made by using higher input resolution or incorporating feature pyramid networks that better preserve fine details.

Q9. Focal Loss in Autonomous Driving Dataset

Topic: RetinaNet & Focal Loss

Difficulty: Hard

Points: 7 marks

SCENARIO

An autonomous vehicle team trains a one-stage detector on a dataset with 200,000 background patches and 3,000 pedestrian instances. Training converges quickly but the model barely outperforms a trivial 'predict background everywhere' baseline. Using standard cross-entropy, training loss reaches ~0.03 within 5 epochs.

TASK

Explain precisely why standard cross-entropy fails here despite low training loss. Define Focal Loss mathematically, explain what each term does, and show qualitatively how it changes the gradient contribution of easy negatives vs. hard positives. What value of γ would you start with and why?

Hint: Consider what dominates the loss sum when 98.5% of samples are easy negatives.

The failure of standard cross entropy loss in this scenario is due to extreme class imbalance between background and object instances. In the dataset most of the samples are easy background regions while only a small fraction correspond to actual objects such as pedestrians.

During training cross entropy treats all samples equally so the large number of easy negative examples dominates the loss. The model quickly learns to classify most regions as background which leads to a low overall loss but does not improve its ability to detect hard positive examples.

This is why the loss decreases rapidly but performance does not improve significantly beyond a simple baseline. Focal loss addresses this problem by down weighting easy examples and focusing training on hard misclassified samples.

It introduces a modulating factor that reduces the contribution of well classified examples so that the model pays more attention to difficult cases. It also includes a balancing term to handle class imbalance explicitly.

As a result the gradient contribution from easy negatives becomes very small while hard positives and difficult negatives receive more emphasis during training. This shifts the learning process toward improving detection performance rather than minimizing loss dominated by trivial examples.

A moderate value of the focusing parameter is typically chosen to balance stability and effectiveness so that the model does not ignore too many examples while still emphasizing hard cases.

Q10. Anchor Box Design for Aerial Object Detection

Topic: R-CNN Family / SSD

Difficulty: Hard

Points: 7 marks

SCENARIO

Your team inherits a Faster R-CNN model pre-trained on COCO. You're adapting it for drone-based infrastructure inspection, detecting objects like power line insulators (very thin, horizontal, ~200:1 aspect ratio) and junction boxes (square, small). COCO anchor boxes don't match these shapes. The model achieves 9% mAP on your test set despite fine-tuning.

TASK

Diagnose why COCO anchor boxes are mismatched for this domain. Describe a principled approach (with a specific clustering method) to design anchors for your dataset. Explain how IoU threshold choice during anchor assignment affects recall vs. precision on thin objects.

Hint: Think about what anchor box shapes COCO was designed around and how they differ from 200:1 aspect ratios.

The poor performance comes from a mismatch between the anchor boxes learned from the coco dataset and the actual shapes of objects in aerial imagery. Coco contains mostly everyday objects with moderate aspect ratios such as people cars and animals so the anchor boxes are designed around roughly square or slightly rectangular shapes.

In aerial infrastructure imagery objects like transmission lines and insulators are extremely elongated with aspect ratios that can reach two hundred to one.

Because of this mismatch the predefined anchors cannot overlap well with ground truth boxes which leads to low overlap during training and poor localization. As a result the model struggles to assign positive matches and fails to learn accurate bounding boxes. A principled way to fix this is to redesign anchor boxes using clustering on the bounding boxes in the target dataset.

A common approach is to apply k means clustering using a distance measure based on overlap rather than euclidean distance so that the resulting anchor shapes reflect the true distribution of object sizes and aspect ratios.

This produces anchors that better match elongated objects and improves assignment quality. The choice of overlap threshold during assignment is also important because very thin objects can have low overlap even when the prediction is reasonable.

If the threshold is too high many valid predictions will be treated as negatives which hurts training. Lowering the threshold slightly allows more matches and improves recall for thin objects while still maintaining reasonable precision. Therefore both anchor redesign and careful threshold selection are necessary to adapt the model to this domain.

Q11. NAS Latency vs. Accuracy in Mobile Deployment

Topic: EfficientNet & NAS

Difficulty: Medium

Points: 6 marks

SCENARIO

A fintech startup used Neural Architecture Search to find an optimal model for fraud detection from transaction images (receipt photos). The NAS process optimized purely for validation accuracy. The winning architecture achieves 97.3% accuracy but has an inference latency of 340ms on their target Android device. Their SLA requires <100ms.

TASK

Explain what NAS is and why optimizing only for accuracy during search produces a latency-violating architecture. Describe how hardware-aware NAS (e.g., MNasNet approach) would have solved this. What proxy metric would you substitute or add to the NAS search objective?

Hint: FLOPs $\neq$ latency. Why does a low-FLOP model sometimes run slower on real hardware than a higher-FLOP one?

Neural architecture search is a method that automatically explores different network designs to find an optimal architecture for a given objective such as accuracy.

In this case the search process was focused only on maximizing validation accuracy without considering how fast the model would run on the target device. This leads to architectures that are computationally heavy or inefficient for the specific hardware even if they perform well in terms of accuracy.

Latency on real devices depends not only on the number of operations but also on memory access patterns parallelism and hardware specific optimizations. This is why a model with fewer operations can sometimes run slower than a model with more operations if it is not well suited to the hardware.

A hardware aware search would include latency as part of the objective function so that the search process balances accuracy with execution time. This can be done by measuring or estimating the inference time of candidate models directly on the target device and incorporating it into the optimization goal.

Instead of using only operation count as a proxy the search should use real latency measurements or a latency predictor trained for that device. By doing so the resulting architecture would satisfy both accuracy and latency constraints and meet the requirement of running within one hundred milliseconds.

Q12. EfficientDet Scaling for Real-Time Endoscopy

Topic: EfficientDet

Difficulty: Medium

Points: 6 marks

SCENARIO

A medical device company needs to detect polyps in colonoscopy video at 25 fps on an NVIDIA Jetson Xavier NX (embedded GPU). They benchmarked EfficientDet-D0 through D7 and found D3 achieves the best accuracy their gastroenterologist approves of, but at 18 fps. D2 runs at 27 fps but misses ~12% of polyps the doctor flags.

TASK

Explain the BiFPN mechanism in EfficientDet and why it is more effective than a standard FPN for this detection task. Given the D2 vs. D3 tradeoff, propose a strategy (architectural, not just hardware) to push D3 toward 25 fps without retraining from scratch.

Hint: What does BiFPN do differently from a simple top-down FPN? Consider inference optimization techniques orthogonal to model size.

The bidirectional feature pyramid network improves upon a standard feature pyramid by allowing information to flow both from high level semantic layers to low level spatial layers and back again through repeated fusion.

In a standard pyramid features are passed in one direction which limits how well different scales interact. The bidirectional design introduces weighted connections that learn how much importance to assign to features from different levels and repeatedly refines them.

This leads to better representation of objects across scales which is especially important in medical imaging where small details such as polyps must be detected accurately. In the given scenario the deeper model achieves higher accuracy but runs slower while the shallower model meets the speed requirement but misses important detections.

A practical strategy is to start from the more accurate model and reduce its computational cost without retraining from scratch. This can be done by simplifying the feature fusion layers reducing the number of channels in the pyramid or removing less important layers based on their contribution.

Another approach is to lower the input resolution slightly while keeping the same architecture so that inference becomes faster with minimal loss in accuracy. Fine tuning the modified model rather than training from scratch allows it to retain most of its learned knowledge while meeting the real time constraint.

Q13. RetinaNet Class Imbalance in Pathology Slides

Topic: RetinaNet & Focal Loss

Difficulty: Hard

Points: 7 marks

SCENARIO

A computational pathology team applies RetinaNet to detect mitotic figures (cell division events) in whole-slide histology images at 40× magnification. Images are tiled into 512×512 patches. Across 1M tiles, only ~0.3% contain a mitotic figure. They find that even with Focal Loss ($\gamma=2$), the model has high precision but extremely low recall (0.41).

TASK

Diagnose why Focal Loss alone is insufficient in this extreme imbalance setting (0.3% positive tiles). Propose two complementary strategies: one at the data/sampling level and one at the loss function level (beyond just increasing γ). Justify your choices with reference to the gradient dynamics.

Hint: Focal Loss was designed for anchor-level imbalance (~1000:1). What is the imbalance ratio here and what does that imply?

The main issue is that focal loss alone is not sufficient when the imbalance exists at the image or tile level rather than only at the anchor level. In this case only about zero point three percent of tiles contain any positive instance which means the imbalance is not just between positive and negative anchors but between entire tiles that contain no useful signal and a very small number that contain objects.

Focal loss was originally designed to handle situations where there are many easy negative anchors within an image but still a reasonable number of positive samples per batch. Here most batches are dominated by completely negative tiles so even after down weighting easy negatives the gradients are still overwhelmed by background examples.

This leads the model to become very conservative and only predict positives when it is highly confident which explains the high precision but very low recall.

The gradient contribution from positive examples is too small and too infrequent to properly shape the decision boundary. At the data level a more effective strategy is to use balanced sampling where each training batch is constructed to include a higher proportion of tiles that contain mitotic figures.

This ensures that positive gradients appear consistently during training and prevents the model from collapsing into a background biased solution.

At the loss level one can introduce class balanced weighting or reweight the loss based on the effective number of samples so that rare positive examples contribute more strongly to the gradient.

This increases the influence of positive samples beyond what focal loss alone provides and helps the model learn to detect harder cases. Together these approaches address both the sampling imbalance and the gradient imbalance which improves recall.

Q14. 1×1 Convolution as Bottleneck in SqueezeNet

Topic: MobileNet & SqueezeNet

Difficulty: Medium

Points: 6 marks

SCENARIO

An IoT sensor startup is deploying SqueezeNet to classify hand gestures on a microcontroller with 256KB flash storage. An intern modified the Fire modules by replacing all 1×1 squeeze layers with 3×3 convolutions to 'capture more spatial context.' The model size jumped from 1.24MB to 6.8MB, breaking the flash constraint.

TASK

Explain exactly why the 1×1 convolution in a Fire module is architecturally essential beyond just parameter reduction — specifically, what role does it serve in the information flow through channels. Describe the mathematical relationship between the squeeze ratio and the channel bottleneck. How would you redesign the Fire module to recover spatial context without the 5× size increase?

Hint: A 1×1 conv is not just a 'small filter' — think about what it computes across the channel dimension.

The one by one convolution in a fire module is essential not only because it reduces parameters but because it controls how information flows across channels. A one by one convolution performs a linear combination across the channel dimension at each spatial location which means it mixes information from different feature maps and creates a compact representation before passing it to the expand layers.

This operation acts as a channel wise projection that selects and compresses the most important features. The squeeze ratio determines how aggressively the number of channels is reduced before expansion. If the input has a certain number of channels and the squeeze layer reduces it to a smaller number then the expand layers operate on this reduced representation which limits the number of parameters in subsequent convolutions.

Mathematically the number of parameters in the expand stage depends on the number of squeeze channels so reducing this number directly reduces overall model size. Removing the one by one convolution and replacing it with a three by three convolution removes this bottleneck and significantly increases both parameters and computation because spatial filtering is now applied before reducing channel dimensionality.

To recover spatial context without increasing model size dramatically one can keep the one by one squeeze layer and add a mix of one by one and three by three filters in the expand stage as originally designed or introduce depthwise separable convolutions in place of standard three by three filters.

This allows spatial information to be captured efficiently while preserving the compact channel representation created by the squeeze layer.

Q15. Full System Design: Smart City Traffic Monitor

Topic: Object Detection (Integration)	Difficulty: Expert	Points: 10 marks
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SCENARIO

A smart city government wants to deploy a traffic monitoring system across 200 intersections. Requirements: (1) detect vehicles, cyclists, and pedestrians in real-time at 30 fps; (2) each edge node has a Jetson Orin with 16GB RAM; (3) the system must handle night, rain, fog, and direct sun glare; (4) privacy law prohibits storing identifiable face/license plate data; (5) budget limits model updates to once per quarter.

TASK

Design the complete detection architecture stack. Choose and justify your backbone (from the architectures studied), detection head, and training strategy. Address: (a) how you handle the multi-condition robustness requirement; (b) what detection architecture you'd use and why (YOLO vs. EfficientDet vs. RetinaNet); (c) how you'd handle the privacy constraint at inference time; (d) how quarterly-update cadence affects your choice of model capacity and fine-tuning approach.

Hint: There is no single correct answer — you are graded on the coherence and depth of your

A complete system design for a smart city traffic monitor must balance accuracy speed robustness and privacy constraints. For the backbone a modern efficient architecture such as an efficientnet based backbone or a lightweight residual network is suitable because it provides strong feature extraction while maintaining computational efficiency on edge hardware.

For the detection head a one stage detector such as a recent yolo variant is preferred because it offers real time performance at thirty frames per second which is required for live traffic monitoring. Compared to two stage detectors it has lower latency and simpler deployment while still achieving good accuracy.

To handle varying environmental conditions such as night rain fog and glare the training strategy should include extensive data augmentation that simulates these conditions along with a diverse dataset collected from different times of day and weather scenarios.

Techniques such as brightness variation noise injection and contrast changes help the model generalize better. For privacy constraints the system should avoid storing raw images and instead process frames in memory and output only anonymized detections such as bounding boxes and object counts.

Additional steps such as blurring or masking faces and license plates at inference time can be applied before any temporary storage or transmission.

The quarterly update cycle means that the model should be stable and not require frequent retraining so a moderately sized model that generalizes well is preferred over a very large one that may overfit quickly.

Fine tuning can be performed periodically using newly collected data to adapt to changes in traffic patterns or environment without retraining the entire model from scratch. This ensures the system remains accurate while respecting deployment and maintenance constraints.

